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Performance-based assistance control for robot-mediated upper-limbs rehabilitation☆

Leigang Zhang [a,*], Shuai Guo [a,b,*], Fengfeng Xi [c]

[a] School of Mechatronic Engineering and Automation, Shanghai University, Shanghai 200444, China
[b] National Demonstration Center for Experiment Engineering Training Education, Shanghai University, Shanghai 200444, China
[c] Department of Aerospace Engineering, Ryerson University, 350 Victoria Street, Toronto, M5B 2K3 ON, Canada

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ABSTRACT

Clinical research results demonstrated that robot-mediated training for the recovery of subjects after the stroke had the benefit of being able to carry out high-intensity repetitive and task-oriented tasks, which could effectively facilitate the rehabilitation effect. However, these subjects may have different training requirements as the disability levels vary from subject to subject at different stages of rehabilitation. Therefore, to meet the rehabilitation training needs of subjects in different rehabilitation stages, a performance-based assistance control strategy for safe robot-mediated therapy is proposed in this paper. The strategy contains three training modes that are switched by the trajectory tracking error, the zero interaction force mode (ZIF), assist-as-needed (AAN) mode, and restriction interaction region (RIR) mode. In ZIF mode, the subject can freely refer to their motion trajectory, and the robot will only offer fixed assistance only when the subject is slacking. In AAN mode, the robot would offer a variable level of assistance force by adjusting the stiffness coefficient to assist subjects in accordance with their motor abilities. To assure safety, a sizable assisting force is applied in RIR mode. To validate and evaluate the effectiveness and performance of the proposed assistance control strategy, a set of task-oriented continuous movement experiments based on an end-effector robot-mediated training system were performed by five able-bodied subjects. The preliminary experiment results demonstrated that the proposed control strategy works properly and could provide corresponding assistance and adjust the working modes according to the subject's motor performance. It was found that the robot provided more assistance as the increasing trajectory tracking error led to a decrease in motor performance. Additionally, adaptive adjustment of the stiffness coefficient allowed the proposed assistance control strategy to induce more active effort. It's worth noting that the parameters of the proposed strategy may be personalized and adjusted for different subjects to satisfy various requirements.

1. Introduction

Subjects after stroke are frequently associated with varying degrees of upper limb motor dysfunction [1], which has a negative impact on their sports ability and activities of daily living and places a financial and emotional burden on them and their family. According to the motor relearning program theory, sufficient rehabilitation training can help stroke survivors regain a part of the upper limb movement ability with careful guidance [2]. Compared with the traditional rehabilitation training method, the robot-mediated method, as a new clinical rehabilitation intervention technique, offers the benefit of producing high-density repeated and task-oriented rehabilitation training [3]. Moreover, clinical study results showed that the robot-mediated training methods have positive effects on the recovery of arm motor function [4, 5].

The mechanical structure of a robot-mediated system plays an important role during robot-mediated physical training. Numerous robot-mediated training systems have been developed over the past few decades, which mainly fall into two categories [6, 7]: the exoskeleton type, such as the Armin system [8] and the Harmony system [9], and the end-effector type, such as the MIT-MANUS system [10] and the GENTLE/S system [11]. These robot-mediated training systems have been

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* Corresponding authors.
E-mail addresses: lgzhang@shu.edu.cn (L. Zhang), guoshuai@shu.edu.cn (S. Guo).

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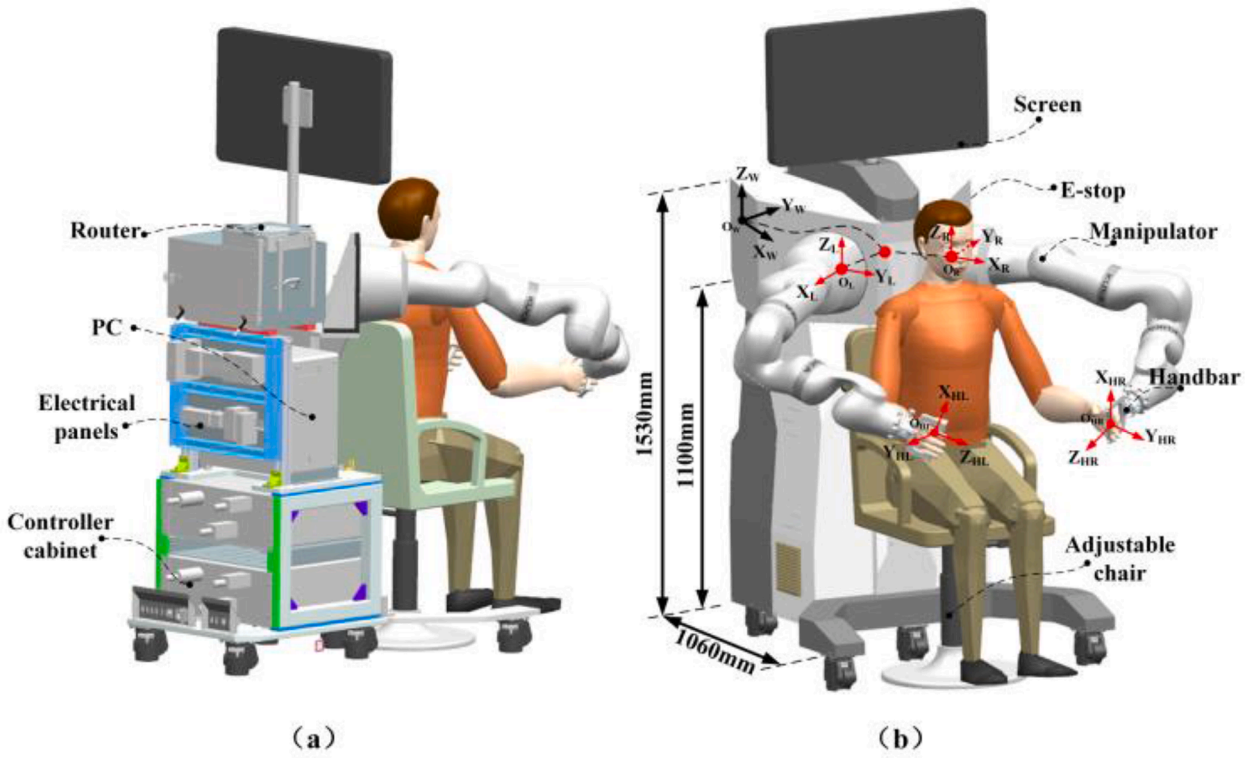


Fig. 1. The mechanical structure of the robot-mediated training system. (a) Schematic diagram of the internal structure. (b) The designed three-dimensional model.

effectively used in clinical rehabilitation therapy and can offer patients sufficient assistance. Although the exoskeleton-type robot-mediated system could carry out rehabilitation training of the single joint of the upper limb, compared with the traditional rehabilitation training methods, the end-effector-type robot-mediated system also has positive effects and superiority in rehabilitation training [12, 13]. Moreover, the end-effector robot-mediated training systems offer some advantages, such as their simple structure and simplified training task planning and adaption to different subjects.

Since robot-mediated training is a patient-centered method that necessitates contact between the patient and the robot, the control strategy of these systems also plays a critical portion during rehabilitation therapy. Recently, some control strategies, such as impedance control [14], Proportion-Integration-Derivative (PID) control [15], have been adopted to assist the upper limb to complete the predetermined training tasks in the robot-mediated system, and the classification and characteristics of these strategies can be found in prior review papers [16–19]. For people who have significant arm motor impairment, the majority of these controllers are designed to rigorously track the desired motion pattern. These control strategies take active interaction forces or moments from the subject as external disturbances and apply opposing control forces to eliminate tracking errors to get effective trajectory tracking outcomes. A stiffer position controller is needed to correctly follow the trajectory because individuals with upper limb motor dysfunction in the early stages of recovery need total support from the robot-mediated system to accomplish the scheduled training activity. However, the active participation of the subjects is considered a critical element that contributes to motor recovery based on the sensorimotor integration theory [20]. Research results show that the minimization of robot-mediated and the maximization of active participation of the subjects during the physical training could stimulate and promote the brain neuroplasticity of patients [21, 22]. Therefore, a control strategy is required to provide necessary assistance or correction to complete rehabilitation training tasks when the subjects are as much as needed. The use of robotic assistance depending on the subject's motor performance, or assist-as-needed (AAN) control, is one method for achieving this characteristic [23, 24]. The AAN strategy can not only stimulate the active participation of subjects during the training tasks but also provoke motor recovery and physical therapy [25, 26]. Research on the AAN control strategy has already been adopted successfully in rehabilitation training of upper limbs [27, 28], lower limbs [29, 30], and fingers [31] to stimulate the subject's active participation.

However, fixed robot assistance cannot be appropriate for subjects at various recovery stages, and the control strategy should be able to regulate the assistance level in accordance with the subject's motor performance. Moreover, the application of control strategy in accordance to the AAN principle in end-effector-type robot-mediated training systems is rare.

To meet the rehabilitation training needs of subjects in different rehabilitation stages, this paper presents an assistance control strategy based on the subject's motor performance to assist the subject's upper limb in performing training tasks for a robot-mediated physical training system. Additionally, three training modes were included and switched in the proposed strategy based on the trajectory tracking error to dynamically and adaptively adjust the assistance level. Taking into account this context, this work presents two main contributions to the field of robot-assisted upper limb rehabilitation:

(1) The robot-mediated assisted control strategy proposed in this paper contains three different training modes: ZIF mode, ANN mode, and RIR mode, which can meet the rehabilitation training needs of subjects in different rehabilitation stages using one controller. Few control strategies can cover the training needs of subjects in any rehabilitation phase.
(2) Safe switching between different rehabilitation training modes during robot-assisted training in the proposed assisted control strategy was achieved by tracking the motion performance of the subject's arm.

The remainder of this paper is organized as follows: the description

and modeling of an end-effector robot-mediated system are introduced in Section 2, and a performance-based assistance control strategy was proposed in Section 3. Experiments with five able-bodied subjects were conducted to evaluate and validate the performance of the proposed strategy in Section 4. Finally, Sections 5 and 6 offer the experiment results, followed by discussions and conclusions.

2. Modeling of the robot-mediated system

The mechanical structure and overall perspective of the robot-mediated training system used in this study are illustrated in Fig. 1, which is an improved version of the system developed in earlier research [32, 33]. The training system is primarily made up of two industrial manipulators (KUKA LBR iiwa7) with seven degrees of freedom (DOF), which are primarily utilized for guiding the upper limb during training, and support for the body structure. The interaction between the subject and the robot adopts a back-to-back mode, which is what sets this system apart from previous robot-mediated systems. Additionally, in this interactive mode, the system's reachability and dexterous workspace may entirely match the physiological reachability and dexterous workspace of the human upper limb [34].

The intersection of the joint axes of two manipulators is selected as the global reference coordinate system to facilitate the ability to describe how the robot moves. The kinematics equation may be expressed as follows, using the right manipulator as an example:

$$\mathbf{T}_{\mathrm{HR}}^{\mathrm{W}} = \mathbf{T}_{0}^{\mathrm{W}} \prod_{i=1}^{7} \left(\mathbf{T}_{i}^{i-1}(q_i)\right) \mathbf{T}_{\mathrm{HR}}^{7} \tag{1}$$

herein, $\mathbf{T}_{\mathrm{HR}}^{7}$ and $\mathbf{T}_{0}^{\mathrm{W}}$ are the transformation matrix between handle and robot flange and the transformation matrix between robot base and global frame, respectively; $\mathbf{T}_{i}^{i-1}$is the transformation matrix between two adjacent links of the manipulator, $\mathbf{q}_i$ is the joint position vector of the manipulator. In general, the nonlinear dynamics equation of the manipulator can be expressed in the joint space as:

$$\mathbf{M}(q)\ddot{\mathbf{q}} + \mathbf{C}(q,\dot{q})\dot{\mathbf{q}} + \mathbf{G}(q) + \mathbf{F}(\dot{q}) = \tau_{\mathrm{c}} + \tau_{\mathrm{ext}} \tag{2}$$

$$\tau_{\mathrm{ext}} = \mathbf{J}^{\mathrm{T}}(q)\mathbf{F}_{\mathrm{ext}} \tag{3}$$

$$\mathbf{J}(q) = \begin{bmatrix} \mathbf{J}_x(q) \\ \mathbf{J}_\theta(q) \end{bmatrix} \tag{4}$$

where $\dot{\mathbf{q}}$and $\ddot{\mathbf{q}}$are joint velocity and acceleration vector of the manipulator, $\mathbf{J}(q)$is the analytic Jacobian matrix and $\mathbf{J}_x(q)$ maps the joint velocity $\dot{\mathbf{q}}$ to translational end-effector velocity and $\mathbf{J}_\theta(q)$ maps the $\dot{\mathbf{q}}$ to rotational end-effector velocity. τ_{c} are the applied torques by joints, $\mathbf{M}(q)$ is the inertial matrix, $\mathbf{C}(q,\dot{q})$ denotes the centrifugal and Coriolis terms, and $\mathbf{G}(q)$ represents the gravity terms, $\mathbf{F}(\dot{q})$ denotes the friction vector, $\mathbf{F}_{ext}$ is an external force applied on the manipulator end-effector, and τ_{ext} is external torque caused by the external force.

Furthermore, the dynamics equation of the manipulator based on the Cartesian variables can be described as:

$$\mathbf{M}_x(x)\ddot{\mathbf{x}} + \mathbf{C}_x(x,\dot{x})\dot{\mathbf{x}} + \mathbf{J}^{-\mathrm{T}}(q)\mathbf{G}(q) + \mathbf{J}^{-\mathrm{T}}(q)\mathbf{F}(\dot{q}) = \mathbf{F}_\tau + \mathbf{F}_{ext} \tag{5}$$

$$\mathbf{M}_x(x) = \left(\mathbf{J}(q)\mathbf{M}(q)^{-1}\mathbf{J}^T(q)\right)^{-1} \tag{6}$$

$$\mathbf{C}_x(x,\dot{x})\dot{x} = \mathbf{M}_x(x)\left(\mathbf{J}(q)\mathbf{M}^{-1}(q)\mathbf{C}(q,\dot{q}) - \dot{\mathbf{J}}(q)\right)\dot{\mathbf{q}} \tag{7}$$

where $\dot{\mathbf{x}}$ and $\ddot{\mathbf{x}}$ are the Cartesian velocity and acceleration vector of the manipulator end-effector, $\mathbf{M}_x$is the end-effector inertial matrix, $\mathbf{C}_x$is the Cartesian Coriolis matrix, and $\mathbf{F}_\tau$,$\mathbf{F}_{ext}$are the control force, and external force vectors, respectively.

Notably, only the end-effector dynamics are considered in (5), the null-space dynamics are not included. However, the manipulator in the paper's robot-mediated training system includes seven DOFs and one redundant kinematic axis (also known as the null space). A straightforward strategy, the null-space projection approach [35], is used in this study to control and regulate the null-space movement. The desired behavior in the robot's null space is characterized in the joint space by the symmetric and positive definite matrices $\mathbf{K}_q$ and $\mathbf{B}_q$ represented, and then projected through a projection matrix $\mathbf{N}(q)$:

$$\tau_{\mathrm{Null}} = \mathbf{N}(q)\left[\mathbf{K}_q\left(\mathbf{q}_{\mathrm{d},0} - \mathbf{q}\right) - \mathbf{B}_q\dot{\mathbf{q}}\right] \tag{8}$$

where the mapping matrix $\mathbf{N}(q)$ is defined as follows:

$$N(q) = I - J^{\mathrm{T}}(q)\left(J^{\mathrm{T}}(q)\right)^{\#} \tag{9}$$

$$\mathbf{J}^{\#}(q) = \mathbf{W}^{-1}\mathbf{J}^{\mathrm{T}}\left(q\right)\left(\mathbf{J}(q)\mathbf{W}^{-1}\mathbf{J}^{\mathrm{T}}(q)\right)^{-1} \tag{10}$$

where $\mathbf{q}_{\mathrm{d},0} \in \mathbb{R}^{7\times 1}$ is the initial pose of the robot, $\mathbf{W}$ and is the weighting matrix. Generally, the weighting matrix $\mathbf{W}$ can be constructed in a variety of ways [36]. The weight matrix $\mathbf{W}$ adopted in this work is a more commonly used least-norm[37] solution $\mathbf{W} = \mathbf{I}$.

3. Performance-based assistance control

The primary focus of this section is the introduction of a performance-based assistance control method that consists of three training modes that may be switched depending on the subject's motor performance.

3.1. Cartesian impedance control

To achieve the compliance interaction between the robot and the subject during the robot-mediated training, the robot needs to be conducted under an impedance control scheme [14]. The idea behind impedance control is to forgo explicitly controlling a specific manipulator variable (e.g., endpoint force, position, and velocity), in the interest of shaping the relationship between input flow variables (e.g., manipulator endpoint position or velocity relative to a reference trajectory) and output effort variables (e.g. net endpoint forces and torques) at an interaction point with the environment. The desired dynamic relationship of the Cartesian impedance control can be expressed as:

$$\mathbf{M}_x\Delta\ddot{\mathbf{x}} + \mathbf{D}_x\Delta\dot{\mathbf{x}} + \mathbf{K}_x\Delta\mathbf{x} = \mathbf{F}_{ext} \tag{11}$$

where $\Delta\mathbf{x} = \mathbf{x}_{\mathrm{cur}} - \mathbf{x}_{\mathrm{des}}$ is the position error between the robot position $\mathbf{x}_{\mathrm{cur}}$ and the desired virtual equilibrium position $\mathbf{x}_{\mathrm{des}}$, and $\mathbf{M}_x$, $\mathbf{D}_x$, and $\mathbf{K}_x$ are the symmetric and positive definite matrices of the desired inertia, damping, and stiffness, respectively. Therefore, the control law of the manipulator in joint space can be obtained:

$$\mathbf{u} = \ddot{\mathbf{q}}_{\mathrm{cur}} = \mathbf{J}^{-1}(q)\left[\ddot{\mathbf{x}}_{des} + \mathbf{M}_x{}^{-1}(\mathbf{F}_{ext} - \mathbf{D}_x\Delta\dot{\mathbf{x}} - \mathbf{K}_x\Delta\mathbf{x}) - \dot{\mathbf{J}}(q)\dot{\mathbf{q}}\right] \tag{12}$$

To simplify and eliminate the external force measurement, the expected inertia matrix of the robot in cartesian space is set identically to the manipulator end-effector inertial matrix $\mathbf{M}_x = \mathbf{M}_x(x)$. In addition, to ensure the passive stability of the system with a time-varying inertial matrix, the Coriolis matrix must be included. The stability proof of that argument is developed in detail in [38], therefore, Eq. (11) can be rewritten as:

$$\mathbf{M}_x(x)\Delta\ddot{\mathbf{x}} + (\mathbf{C}_x(x,\dot{x}) + \mathbf{D}_x)\Delta\dot{\mathbf{x}} + \mathbf{K}_x\Delta\mathbf{x} = \mathbf{F}_{ext} \tag{13}$$

Since the inertia matrix in (13) depends on the end-effector pose, the damping matrix should not be a constant matrix for the same reasons as mentioned above. Therefore, a robot pose-dependent damping matrix is adopted [39], and the desired robot behavior can be achieved by the following control law:

$$\mathbf{F}_{ext} = \mathbf{G}_x(x) + \mathbf{M}_x(x)\ddot{\mathbf{x}}_d + \mathbf{C}_x(x,\dot{x})\dot{\mathbf{x}}_{\mathrm{d}} - \mathbf{D}_x(x)\Delta\dot{\mathbf{x}} - \mathbf{K}_x\Delta\mathbf{x} \tag{14}$$

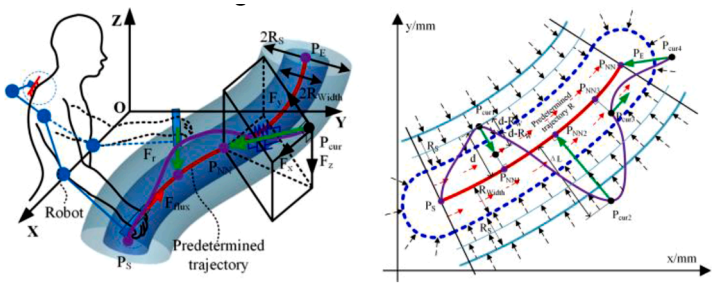


Fig. 2. Schematic diagram of enhancing the spatial virtual ZIR. (a) The schematic diagram of the ZIR. (b) The schematic diagram of $\mathbf{P}_{\mathrm{NN}}$ and the assistance provided by the robot. Ps and Pe are the starts and endpoint of the predetermined trajectory, and the purple and red curve are for the actual and predetermined trajectory respectively.

Finally, the input torque of the Cartesian impedance controller is then defined as:

$$\tau_c = \mathbf{J}^T(q)\left[\mathbf{M}_x(x)\ddot{\mathbf{x}}_{des} + \mathbf{C}_x(x,\dot{x})\dot{\mathbf{x}}_{des} - \mathbf{D}_x(x)\Delta\dot{\mathbf{x}} - \mathbf{K}_x\Delta\mathbf{x}\right] + \mathbf{F}(\dot{q}) + \mathbf{G}(q) \tag{15}$$

3.2. Performance-based assistance control strategy

Impedance control is frequently adopted in robot-mediated rehabilitation therapy by delivering AAN characteristics to make the robot display compliance in Cartesian space and enable the subject to depart from the predetermined trajectory. However, the impedance control strategy does not allow temporal freedom, and the robot will assist as long as the subject deviates from the timed predetermined trajectory. Therefore, a performance-based assistance control strategy that contains three training modes switched through the trajectory error is proposed to promote rehabilitation training.

First, as illustrated in Fig. 2, a deadband Zero Impedance Region (ZIR) was used to replace the predefined trajectory. In this study, the estimator of the subject's motor performance relied on the kinematic deviation between the actual trajectory and the predetermined trajectory. Furthermore, there are three different training modes based on the subject's motion trajectory tracking error $\Delta\mathbf{L}$, which can be defined as:

$$\Delta\mathbf{L} = \mathbf{P}_{\mathrm{cur}} - \mathbf{P}_{\mathrm{NN}}(R) \tag{16}$$

where $\mathbf{P}_{cur}$ and $\mathbf{P}_{NN}$ are the actual position of the subject and the nearest point on the predetermined trajectory, respectively. When the trajectory tracking error of the subject is small $\| \Delta\mathbf{L} \|_2 \in [0, \mathrm{R}_{width}]$, the robot does not provide any assistance within the ZIR. The subject could choose their preferred trajectory, and the robot works in zero interaction force mode (ZIF). As shown in the blue region in Fig. 2, the robot was in a complete backward-drive state within the ZIR with an adjustable radius R_{width}. The subject is deemed to possess sufficient capacity to fulfill the predetermined tasks in this mode. However, the subject might remain in the ZIR instead of actively trying to complete the task, which might get stuck in a position. The tendency or slacking for the subject to rely heavily on the assistance force has been recorded [40], which will greatly reduce the active participation and effort of subjects. In the ZIF mode, a minimum velocity curve is introduced to counteract this, which will build a virtual movement. The robot will produce a moving wall to push the subject to keep up with the virtual movement if they are still or moving at a pace less than the minimum velocity, and the robot provides a minimum assistance force to help to complete the predetermined task and avoid slacking. The minimum assistance force $\mathbf{F}_{flux}$ can be calculated as:

$$\mathbf{F}_{flux} = \begin{cases} k_{flux}\dfrac{\mathbf{t}_{flux}}{|\mathbf{t}_{flux}|} \| \Delta\mathbf{L} \|_2 \in [0, R_{width}] \\ 0 else \end{cases} \tag{17}$$

$$\mathbf{t}_{flux} = \frac{d}{dR}\mathbf{P}_{NN}(R) \tag{18}$$

when the tracking error is increasing $\| \Delta\mathbf{L} \|_2 \in [R_{width}, \mathrm{R}_s]$ (deviating from the ZIR), the assistance mode is switched and the robot now works in AAN mode. In such a situation, the trajectory tracking error of the subject indicates that the training task on the predetermined trajectory cannot be completed. Thus, the robot needs to provide corresponding assistance forces to assist the subject to complete the task, which could prevent the subject from losing confidence because of poor performance. One way to modulate the assistance force provided by the robot is to modify the mechanical impedance in the Cartesian space rendered by the end-effector of the robot. Typically, a common impedance-based AAN control strategy is to update nominalized impedance parameters (including stiffness and damping), and assistance force provided by the robot can be expressed as:

$$\mathbf{F}_{assist} = \left(\mathbf{K}_{x_wall}(\| \Delta\mathbf{L} \|_2 - R_{width}) + \mathbf{D}_{x_wall}\frac{d}{dt}(\| \Delta\mathbf{L} \|_2 - R_{width})\right) \times \vec{\mathbf{n}} \tag{19}$$

$$\vec{\mathbf{n}} = \frac{\mathbf{P}_{cur} - \mathbf{P}_{NN}(R)}{\| \Delta\mathbf{L} \|_2} \tag{20}$$

where $\mathbf{K}_{x_wall}$ and $\mathbf{D}_{x_wall}$ are the positive diagonal stiffness and damping matrix. A large assistance force will be generated even if the trajectory tracking error is small If the stiffness factor $\mathbf{K}_{x_wall}$ in the AAN mode became too large. When the stiffness factor is large enough, the robot will accurately track the predetermined trajectory and ignore the subject's active efforts. However, the assistance provided by the robot with a fixed stiffness factor parameter can not suitable for subjects with various levels of injuries at different recovery stages. In this paper, the assistance force during the robot-mediated physical training is modified by adjusting the mechanical impedance of the robot in Cartesian space, and the impedance coefficient $\Delta\mathbf{K}$ is modified according to the trajectory tracking error of the subject:

$$\mathbf{K}_{x_wall} = \mathbf{K}_{min} + \Delta\mathbf{K} \tag{21}$$

$$\Delta\mathbf{K} = \frac{\mathbf{K}_{\max} - \mathbf{K}_{\min}}{R_s - R_{\mathrm{width}}} \tag{22}$$

where $\mathbf{K}_{min}$ and $\mathbf{K}_{min}$ represent the upper and lower limit of the stiffness factor in AAN mode. In the AAN mode, there is a trade-off between the trajectory tracking error and the subject's active effort. If the trajectory tracking error decreases, a lower impedance coefficient should be

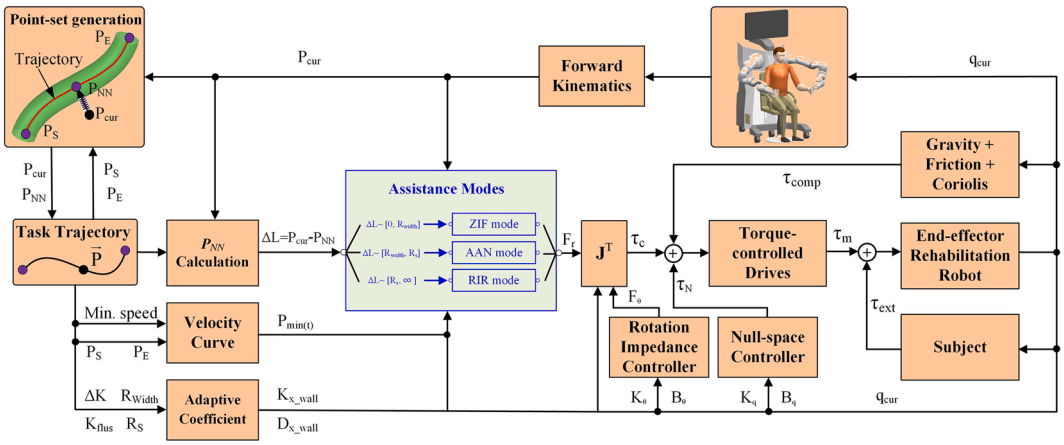


Fig. 3. The overall control structure of the proposed assistance control strategy.

employed to encourage the subject to actively participate and finish the assigned training activities. In contrast, a higher impedance coefficient should be used to assist the subject in successfully reducing the trajectory error as the trajectory tracking error begins to rise. It should be noted that there are different methods to modulate the damping force or the stiffness force since the damping force is velocity-dependent and the stiffness force is position-dependent. However, if the trajectory tracking error is large enough $\| \Delta \mathbf{L} \|_2 \in [R_s, \infty]$, the robot would work in the restriction interaction region (RIR) mode. This indicates that the subject's performance is poor, and the robot will give priority to the safety of the subject. At this time, the robot will exert a large assistance force, which can be calculated as follows:

$$\mathbf{F}_{assist} = (\mathbf{K}_{\max} \cdot (\| \Delta \mathbf{L} \|_2 - R_{width})) \times \vec{\mathbf{n}} \tag{23}$$

If the subject is still unable to decrease the trajectory tracking error with the assistance force provided by the robot, we believe that the training task is difficult for the subject. Therefore, the training task should be stopped and adjusted appropriately. Finally, the robot's assistance force $\mathbf{F}_r$ during the robot-mediated training according to the proposed control strategy and the control torque τ_c can be calculated as follows:

$$\mathbf{F}_r = \mathbf{F}_{assist} + \mathbf{F}_{flux} \tag{24}$$

The robot's end-pose during robot-mediated training, however, cannot be arbitrary due to the physical range of motion (ROM) of the wrist joint since the patient and the robot interact in back-to-back mode. The robot end pose cannot be fully uncontrollable or in a fixed position while taking into account the interaction between the robot end and the subject. Therefore, a Cartesian rotation impedance controller based on Euler angles was adopted to manage the pose at the end of the robot. By choosing an initial Euler angular pose for the robot's end θ_d, a rotational Cartesian impedance controller is applied as below:

$$\mathbf{F}_\theta = \mathbf{K}_\theta(\theta_d - \theta) - \mathbf{B}_\theta \dot{\theta} \tag{25}$$

$\boldsymbol{\tau}_\theta = \mathbf{J}_\theta^{\mathrm{T}}(q)\mathbf{F}_\theta$ (26) where $\mathbf{K}_{\boldsymbol{\theta}} \in \mathbb{R}^{3\times3}$ and $\mathbf{B}_{\boldsymbol{\theta}} \in \mathbb{R}^{3\times3}$ are the impedance parameters of the rotational impedance controller, $\boldsymbol{\theta} \in \mathbb{R}^{3\times1}$ is the Euler angular pose of the robot's end respectively, $\mathbf{K}_{\boldsymbol{\theta}}$ and $\mathbf{B}_{\boldsymbol{\theta}}$ are chosen to approximate a constant damping ratio for each rotational direction in the end-effector impedance controller. The rotation damping factors in the three rotation directions at the end of the robot are approximated as:

$$\varsigma_i = \frac{b_i}{2m_i\sqrt{k_i/m_i}} \tag{27}$$

herein, k_i and b_i represent the diagonal elements of $\mathbf{K}_\theta$ and $\mathbf{B}_\theta$ respectively, and $m_i$is one of the corresponding three diagonal elements of the rotational end-effector mass matrix[41], which is given by (6). The appropriate damping matrix $\mathbf{B}_\theta$ is selected to create a constant damping factor $\varsigma_i \approx 0.4$ based on the stiffness $\mathbf{K}_\theta$ of our chosen rotational impedance controller. This option falls between undamped $\varsigma_i = 0$ and critically damped $\varsigma_i = 1$. As a result, between the robot end and the predetermined trajectory, a 3D rotating spring with stiffness $\mathbf{K}_\theta$ is introduced. This guarantees that the robot's posture deforms during motion assistance and promotes interaction with the subject's arm.

Therefore, the control torque τ to achieve the proposed assistance control strategy can be calculated finally as follows:

$$\tau = \mathbf{J}^{\mathrm{T}}(q)\left[\mathbf{M}_x(x)\ddot{\mathbf{x}}_{\mathrm{des}} + \mathbf{C}_x(x,\dot{x})\dot{\mathbf{x}}_{\mathrm{des}}\right] + \mathbf{F}(\dot{q}) + \mathbf{G}(q) + \mathbf{J}_x^T(q)\mathbf{F}_{\mathrm{r}} + \tau_\theta + \tau_{\mathrm{Null}} \tag{28}$$

The corresponding simplified control flowchart of the proposed performance-based assistance control strategy is shown in Fig. 3. The high-level controller of the control strategy described in the study, which determines the necessary assistance force based on the subject's arm's motion performance, is depicted on the left side in Fig. 3. The low-level controller is located on the right side and is primarily used to follow the commands from the high-level controller while compensating the robot's dynamics model. The three training modes (ZIF, AAN, and RIR) in the control strategy are selected and switched for in the center blue region. Depending on the subject's arm motor performance, the controller will evaluate them and switch to the desired training mode. In addition, our control strategy is built using the robot impedance controller, whose passivity can be assured although it possesses the aforementioned features.

Table 1
Details of five able-bodied subjects recruited for the evaluation experiments.

Subject	Gender	Age	Weight	Height	Impairment history
1	Male	30	68 Kg	1.82 m	No
2	Male	24	65 Kg	1.68 m	No
3	Male	27	69 Kg	1.70 m	No
4	Male	27	62 Kg	1.70 m	No
5	Male	25	64 Kg	1.72 m	No

4. Experimental validation

The purpose of this experiment is to verify and evaluate the performance of the proposed assistance control strategy with continuous tracking movement on the presented robot-mediated system (shown in Fig. 1). Through the Fast Robot Interface (FRI), the manipulator in the robot-mediated system is capable of performing external torque control at up to 1 kHz loop rates [42]. Besides, the FRI friction and gravity compensation of the robot was activated throughout the experiments.

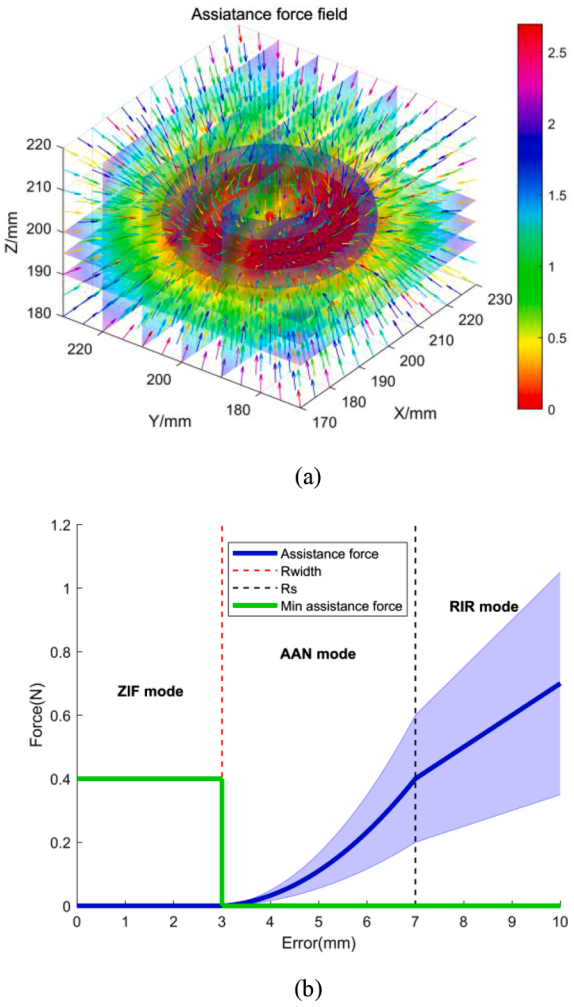


Fig. 4. Simulation results of the three training modes (ZIF mode, AAN mode, RIR mode). (a) The simulation results of the assistance force distribution. (b) Relationship between the assistance force and trajectory tracking error in different modes.

In this experiment, five able-bodied subjects (5 males, average age (mean 26.6 years with a standard deviation (STD) of 2.3, height (1.72 m with STD 0.1) and weight (65.6 Kg with STD 2.9)) were recruited, and none of these subjects had a musculoskeletal or neurological disease. Detailed information of these subjects is shown in Table 1.

We assume that the subject's right arm is used during the experiments. This study was approved by the ethics committee of the Yueyang Hospital of Integrated Traditional Chinese and Western Medicine. Written informed consent was collected from all subjects before participating in this study.

Before the experiment, simulations were carried out to demonstrate and evaluate the assistance control (using a circular trajectory as an example); the results are given in Fig. 4. When the robot is inside the red region, where it has unrestricted movement, it works in ZIF mode. When it is between the red region and the blue region, it operates in AAN mode and RIR mode when it is outside the blue book region. The color distribution in Fig. 4 represents the magnitude assistance force provided by the robot, while the arrows represent the direction of the assistance force. The simulation results above show that if the subject's motion velocity is less than the predetermined minimum velocity, the robot will offer a fixed assistance force in ZIF mode (slacking occurs). In AAN mode, the robot will provide a small assistance force in response to an increase in trajectory tracking error, but in RIR mode, a substantial assistance force will be applied to assure the subject's safety. Some

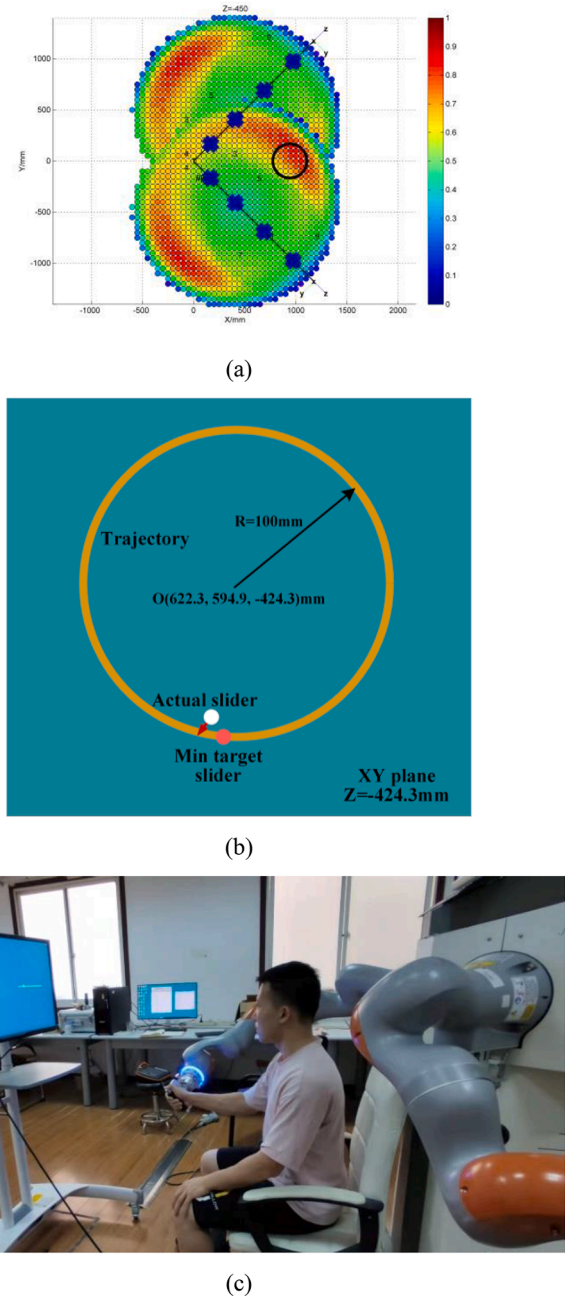


Fig. 5. The chosen training trajectory and unity-based scene. (a) The training trajectory. (b) The visual feedback scene. (c) The experimental test scenario of a part subject.

important parameters for simulation are shown as follows:R_{width} = 30mm, R_S = 70*mm*,$\mathbf{K}_{min}$ = *diag*[10,10,10](N/m),$\Delta\mathbf{K}$ = *diag*[22.5,22.5, 22.5](N/m), $\mathbf{K}_{max}$ = *diag*[100,100,100](N/m), k_{flux} = 0.4N.

Fig. 4(a) shows the distribution of the robot assistance force field. Fig. 4(b) shows the relationship curves between the assistance force and the trajectory tracking error under different training modes. The switch of the three training modes (ZIF, AAN, and RIR) of the assistance control strategy is realized based on the trajectory tracking error of the subject.

These participants were given explicit instructions throughout the experiment to try their best to carry out continuous tracking motions on a circular trajectory in a horizontal plane. The center of the trajectory in this instance and its radius is (622.3 mm, 594.9 mm, −424.3 mm) and 100 mm, respectively. In this work, a circular trajectory was chosen based on the dexterous map of the robot [34, 33] (the black circle in

Table 2
Parameters during the experiments.

Parameters	Value	Units
R_{width}	20	mm
R_S	90	mm
$\mathbf{K}_{min}$	$diag[40, 40, 40]$	N/m
$\mathbf{K}_{max}$	$diag[600, 600, 600]$	N/m
$\mathbf{D}_{x_wall}$	$diag[43.3, 31.68, 37.19]$	N − s/m
$\mathbf{K}_{flux}$	1	N
$\mathbf{K}_{\theta}$	$diag[100, 100, 100]$	N/rad
$\mathbf{B}_{\theta}$	$diag[3.0, 3.4, 1.6]$	N − s/rad
θ_{d}	$[0.4, 0.5, -2.0]$	rad
$\mathbf{K}_q$	$diag[10, 10, 10, 10, 5, 5, 1]$	N/rad
$\mathbf{B}_q$	$diag[2.5, 3.6, 2.1, 2.1, 0.3, 0.2, 0.1]$	N − s/rad
$\mathbf{q}_{\mathrm{d},0}$	$[30, -10, 20, 65, 10, -50, 0]$	deg

Fig. 5(a)), which can be entirely covered within the robot's workspace. Additionally, a Unity-based visual feedback scene that is displayed on a screen in front of the subject has been developed to assist the subjects in feeling the spatial trajectory, as illustrated in Fig. 5(c). The yellow curve in the scene in Fig. 5(b) is for the predetermined trajectory and the red arrow represents the magnitude of the assistance force. The actual and minimum positions (adjustable) were displayed in different colors (white and red) and two-sphere balls indicated the minimum and actual position at every time point. The detailed parameters of the proposed assistance control strategy during the experiments are shown in Table 2.

In situations when able-bodied subjects could do the aforementioned activity and follow the specified trajectory well, the robot would mostly operate in ZIF mode. The participants were asked to follow the trajectory of six revolutions in total to verify and evaluate the impacts of various modalities of the robot-mediated assistance control method. The individuals traveled mostly in the ZIF mode during the first two laps (slacking), pretending to go more slowly than the minimal velocity to imitate slacking (in a relaxed condition). The participants were requested to actively track the trajectory in the second and third laps while being guided by the Unity-based environment (active), whereas they were instructed to purposefully increase the trajectory tracking error in the last two laps to simulate the bad motor performance (poor performance). Before conducting the experiments, the subjects were able to complete a few simple tasks until they became familiar with the robot-mediated physical training system. To ensure participants' safety during robot-mediated training, hardware and software safety protections are implemented. The e-stop outside the body, speed, and interface force overload detection programs in the software are presented, respectively. The whole system could be stopped immediately by a scram button for safety.

In the experiments, the assistance forces and the robot's motion data were collected and calculated respectively. The kinematics equation of the robot may be used to solve the position of the subject's arm since the subject's hand is attached to it via a specially designed handle. The mean absolute error (MAE) between the actual trajectory and predicted trajectory was computed to evaluate the motor performance of the subject's limb, which can be calculated as follows:

$$\mathrm{MAE} = \frac{1}{N}\sum_{j=1}^{N} \| \mathbf{P}_{cur}(j) - \mathbf{P}_{NN}(j) \|_2 \tag{29}$$

where N is the number of sampling points in the predetermined task. Similarly, to further analyze the assistance force provided by the robot, the average assistance force (AAF) was defined as follows:

$$\mathrm{AAF} = \frac{1}{N}\sum_{j=1}^{N} \| \mathbf{F}_{\mathrm{r}}(j) \|_2 \tag{30}$$

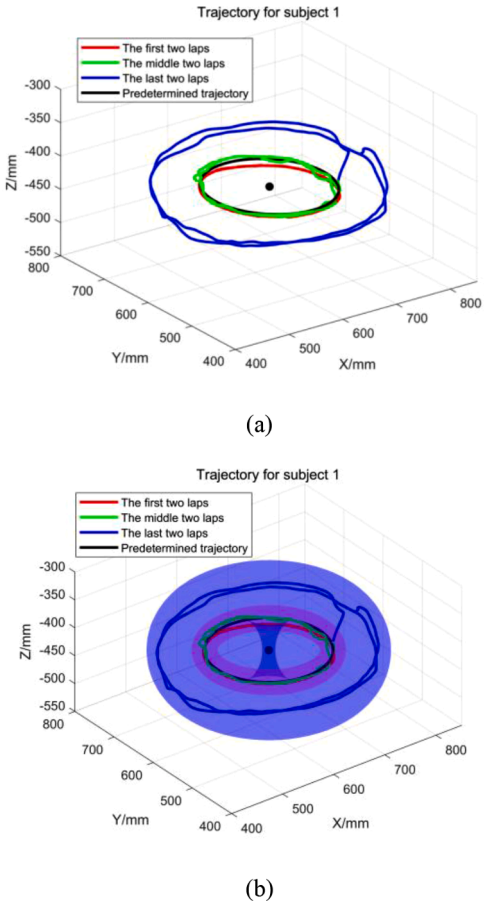


Fig. 6. The trajectory of subject 1 during the experiment. (a) Trajectories. (b) Trajectories with a virtual tunnel.

5. Results and discussion

The red channel and the blue channel in Fig. 6 represent, respectively, the limits of the three modes in the robot-mediated strategy while subject 1 performed the experiment task. The red, green, blue, and black curves are for the actual trajectories of three states during the experiments (slacking, active, poor) and predetermined trajectories respectively. The latter two laps are primarily concentrated between the red channel and the blue channel, according to the experiment procedures, whereas the first four laps are primarily focused on the red channel.

Fig. 8 shows the MAE results of the trajectory tracking error between the actual and predetermined trajectory for five subjects in three different cases during the experiments. The MAE of the five subjects in the first two laps was 7.3 mm, 3.3 mm, 6.5 mm, 2.8 mm, and 2.6 mm, while the MAE in 3–4 laps is 6.2 mm, 5.4 mm, 7.1 mm, 8.7 mm, 5.0 mm, and the MAE in the last two laps were 66.3 mm, 73.1 mm, 53.0 mm, 65.1 mm, and 73.4 mm, respectively.

The boundaries of the switching of the three modes are shown by the red and black dotted lines in Fig. 8, which also shows the relationship between the robot's assistance force and subject 1′s trajectory tracking error. Consistent with the results in Fig. 6, movement in the first four laps was mainly concentrated in the red channel, while the movement in the last two laps was mainly concentrated between the red channel and the blue channel, and RIR mode was triggered in the last two laps (shown as the arrow in Fig. 8). Fig. 9 shows the AAF results of five able-bodied subjects during the experiment. The AAF in the first two laps was 8.2 N, 5.3 N, 7.5 N, 4.7 N, and 4.3 N, while the AAF in 3–4 laps is 1.4 N, 1.3 N, 1.4 N, 1.4 N, 1.3 N, and the AAF in the last two laps was 19.5 N, 26.1 N, 12.7 N, 21.8 N, 26.7 N. respectively.

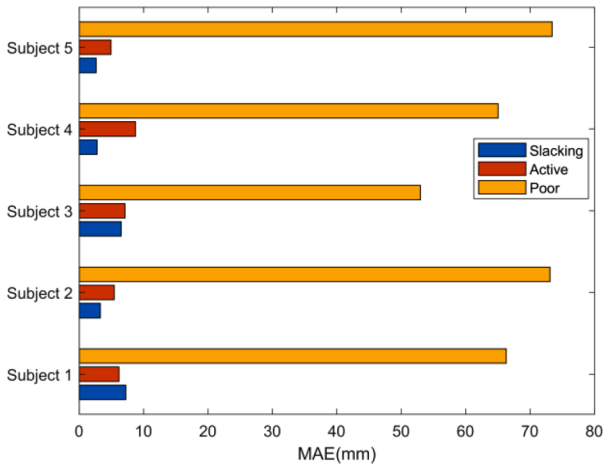


Fig. 7. The MAE results of five able-bodied subjects in different cases.

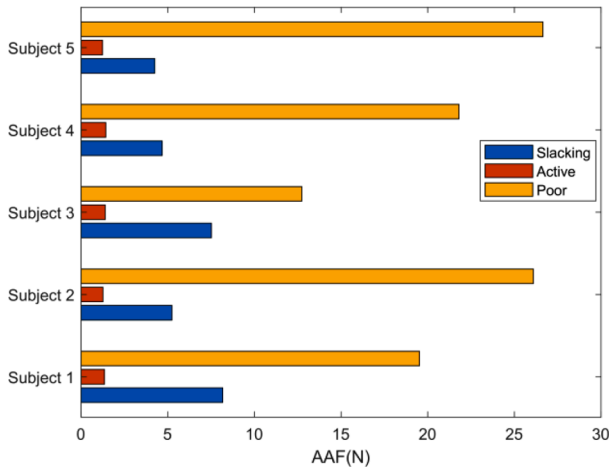


Fig. 9. The AAF results of five able-bodied subjects in different cases.

Fig. 10(a) primarily displays the robot's overall assistance force and its component forces acting on Subject 1 in three directions. The entire interaction moment between subject 1 and the robot end is shown in Fig. 10(b), with an average value of 1.9 N.m.

Fig. 11 primarily displays the mean and standard deviation of the MAE and AAF of five able-bodied subjects throughout the first two, third to fourth, and final two laps, respectively. The mean MAE and AAF of five subjects were 4.5 mm (STD: 2.2 mm) and 5.7 N (STD: 2.2 N) in the first two laps, 6.5 mm (STD: 1.5 mm) and 1.3 N (STD: 0.1 N) in the third and fourth laps, and 66.2 mm (STD: 8.3 mm) and 21.4 N (STD: 5.7 N), respectively, in the final two laps.

This paper presents a robot-mediated training assistance control strategy for continuous tracking movement, which includes three different assistance modes that are switched according to the trajectory tracking error of the subject. Furthermore, the performance of the control strategy was preliminarily validated and evaluated through a series of experiments based on five able-bodied subjects. As shown in Figs. 6 and 8, the movement of the first four laps (slacking and active) is mainly concentrated in the red channel, that is, the robot mainly works in the ZIF mode, while the last two laps mainly (poor) work in the AAN mode and the RIR mode. The robot will provide a fixed assistance force (which was set at 1 N throughout the experiment) while the control strategy is working in ZIF mode and slacking happens to make sure that the subject's movement can catch up with the predefined minimum speed. However, when the subject travels independently in the red channel (shown in Figs. 6 and 8), the subject's trajectory tracking error is minimal and the robot's average assistance force is 1.3 N. The assistance force offered by the robot will grow as the trajectory tracking error increases when the control strategy works in the AAN mode and the subject's trajectory tracking error is between the red channel and the blue channel (shown in Fig. 8). The impedance stiffness coefficient will also vary with the trajectory tracking error. The control strategy works in the RIR mode when the trajectory tracking error is significant enough to diverge from the blue channel (shown in Fig. 8, at the 85 s approximately), and the robot provides a large assistance force to ensure the safety of the subjects.

As can be seen from the experiment results of five able-bodied subjects, when the control strategy works in ZIF mode and slacking occurs (the first two laps), as shown in Figs. 7, 9, and 11, the mean MAE is 4.5 mm, and the mean AAF is 5.7 N. The average assistance force herein is larger than the predetermined value of 1 N, mainly because the subject is in a relaxed state and the subject's arm has a certain gravity, the robot needs to overcome the weight of the arm to perform the predetermined task. As a result, the robot must overcome the weight of the arm to complete the task. The mean MAE and mean AAF are 6.5 mm and 1.3 N, respectively, while the control strategy works in the ZIF mode without slacking (the third and fourth laps). Since the impedance stiffness coefficient in ZIF mode was set at a relatively low value for safety reasons, the assistance force offered by the robot is not strictly equal to zero. The

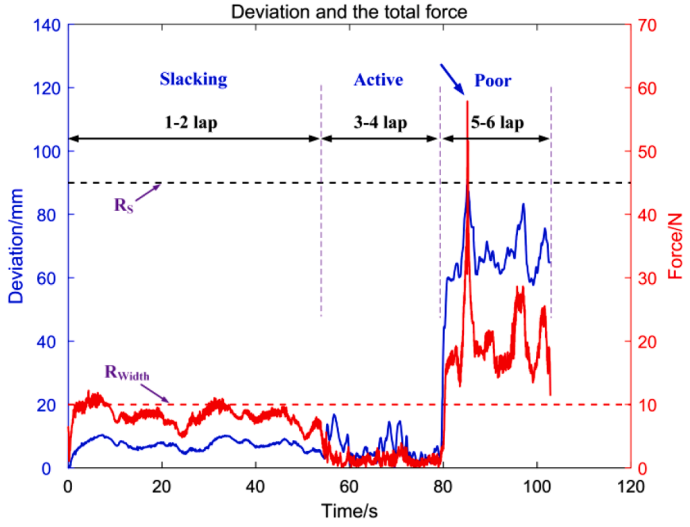


Fig. 8. The total assistance force provided by the robot and the trajectory tracking error of the subject.

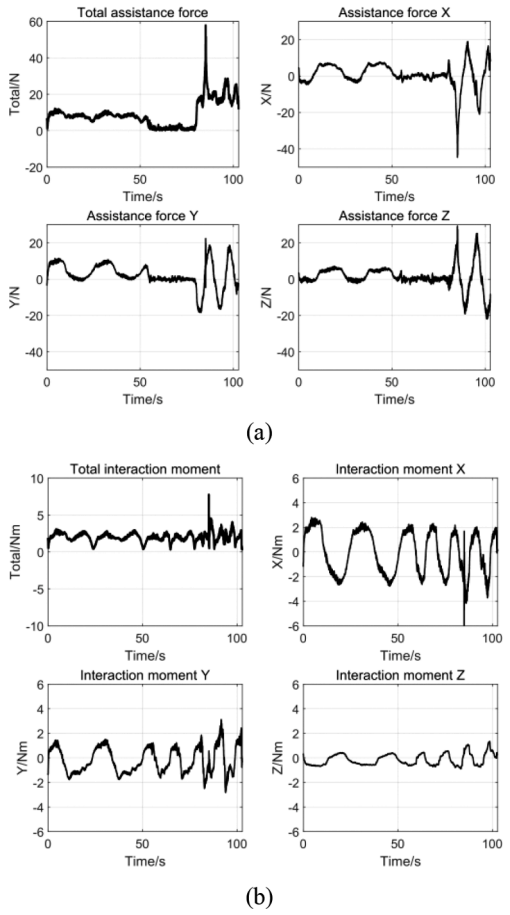


Fig. 10. The assistance force and the interaction moment between the subject and robot. (a) is for the assistance force. (b) is for the interaction moment between the subject and the robot.

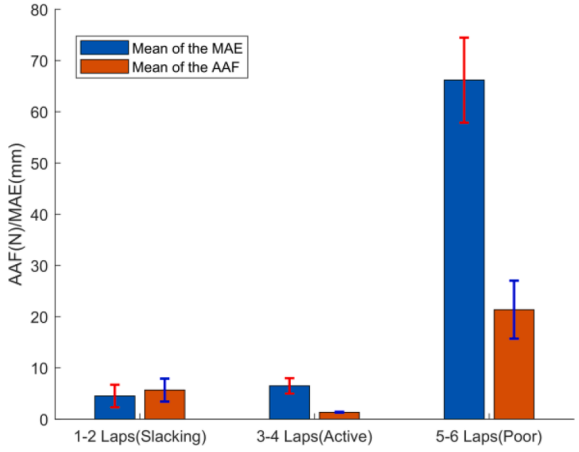


Fig. 11. The mean and standard deviation of the MAE and AAF of five able-bodied subjects.

experiment's results indicated that the mean MAE was 66.2 mm and the mean AAF was 21.4 N when the control strategy works in the AAN mode (the final two laps). As can be observed from the simulation results (shown in Fig. 4(b)) and experiment (shown in Fig. 8) results, the robot modifies the assistance force in accordance with the subject's trajectory tracking error. When the control strategy works in the RIR mode, as shown in Fig. 8 (see the arrow), the robot provides a large assistance force mainly to prevent the safety of the subjects resulting from poor performance. Since the desired pose of the robot of the rotational impedance controller remained unchanged throughout the experiment, the interaction moment between the subjects and the robot was minimal (average value was 1.9 N.m, as shown in Fig. 10(b)).

Traditional AAN control techniques based on end-effector robot-mediated training systems may employ performance-based control strategies or deliver AAN property by monitoring the tracking error in the velocity domain [10, 28, 43]. The assistance control strategy proposed in this paper could provide the subject with spatial and temporal freedom, and automatically switch the training mode of the control strategy according to the subject's motor performance. The fixed assistance coefficient struggles to meet the requirements of rehabilitation training due to the diverse patients' varying levels of injuries [44]. The robot-mediated assistance control strategy proposed in this paper could adjust the stiffness coefficient according to the subject's motor performance. The stiffness factor and the assistance force provided by the robot increase in proportion to the subject's deviation since a bigger deviation indicates worse motor performance. The assistance control strategy presented in this work has configurable parameters in the high-level controller, and the experimental section of the study also assesses the accuracy and viability of the control strategy using a predetermined controller parameter. By changing these factors, the control strategy may be made more appropriate. As we described in the introduction, adjusting the parameters of the control strategy can meet the training needs of different rehabilitation stages.

In scenarios involving human-robot interaction, reasonable modeling of the human-machine coupled dynamics is crucial for the design of the controller [45, 46], which can provide useful insights for the optimization of the controller. However, in the control strategy proposed in the paper, we treat the active effort of the subject as an external force to the robot system and not as part of the robot control system. Moreover, in our proposed assisted control strategy we need the robot to provide appropriate assistance when the motion performance of the subject's arm deteriorates, without fully conforming to the motion of the subject's arm.

There are a few limitations in this study that might be addressed in follow-up investigations. A six-dimensional force sensor might also be built onto the end-effector to increase the measurement of the interaction force. Five healthy volunteers were recruited to take part in the experiment to validate and evaluate the effectiveness of the suggested control technique. It is important to do more research based on a patient-based experiment to demonstrate the effectiveness of the suggested control strategy to advance its clinical application. In addition, subsequent studies should model the human-robot coupling dynamics to facilitate human-robot interaction by using the subject's active effort as part of the controller.

6. Conclusions

Patients after stroke are commonly accompanied by a series of injuries and have various rehabilitation training requirements in different recovery stages. To satisfy such requirements and promote activity effort, a performance-based assistance control strategy for safe robot-mediated therapy is proposed in this paper. To validate and evaluate the performance of the control strategy, five able-bodied subjects were recruited to perform a series of task-oriented experiments based on an end-effector robot-mediated training system. The preliminary experiment results demonstrated that the proposed performance-based assistance control strategy works properly and could switch the training modes well according to the subject's motor performance, and corresponding assistance force would be provided in different modes. The evaluation experiment displays the great potential of the proposed control strategy in clinical robot-mediated rehabilitation training.

As a prospective study, only able-bodied subjects were recruited for the experiment in this work. Nonetheless, in the next stage, experiments with patients-based are needed to verify and evaluate the performance

and efficacy of the proposed strategy and to help to increase the training intensity on the basis of the performance that we have achieved.

CRediT authorship contribution statement

Leigang Zhang: Methodology, Software, Validation, Investigation, Writing – review & editing, Visualization. **Shuai Guo:** Conceptualization, Supervision. **Fengfeng Xi:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mechatronics.2022.102919.

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Leigang Zhang received the B.S and M.S. degrees in mechanical engineering from Anhui University of Science and Technology, Anhui, China, in 2014 and 2017. He is currently pursuing a Ph.D. degree in mechanical engineering and automation from Shanghai University, Shanghai, China. His-current research interests include robotics, impedance control, and upper-limb rehabilitation robot.



Shuai Guo received a Ph.D. degree with a major in mechanical engineering and automation from Shanghai University, Shanghai, China, in 2006. He is currently a professor at Shanghai University and vice president of the Shanghai Robot Industry Technology Research Institute. His-research interests include robotics, rehabilitation robot, and industrial robot.



Fengfeng Xi received the Ph.D degrees in mechanical engineering from University of Toronto, Toronto, Canada, in 1993. He is currently a professor at Ryerson University and the director of the Ryerson Institute for Aerospace Design and Innovation (RIADI). His-research includes computational dynamics, designing smart aircraft cabins and seats, and aircraft manufacturing automation.